

Prime-intersection families: a resolution of Problem 6 and a Granville-type conjecture

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Abstract

Problem 6 in the curated list *Ten New Erdős-style problems in number theory and combinatorics* asks whether the maximum size $P(N)$ of a family $\mathcal{F} \subseteq 2^{[N]}$ with all pairwise intersection sizes prime is polynomial in N , or whether it can be exponential. The dichotomy is false: $P(N)$ is superpolynomial but subexponential. More quantitatively, for an absolute constant $c > 0$,

$$N^{c \log_2^{(7)} N} \leq P(N) \leq \exp\left(\left(\frac{1}{2} + o(1)\right) \frac{N \log \log N}{\log N}\right),$$

where $\log_2^{(7)}$ is the seven-fold iterated base-2 logarithm. The lower bound comes from arithmetic progressions of primes and a block construction. The upper bound is a layer-by-layer polynomial-method argument, sharpened by counting only the primes that can occur in a fixed uniform layer. The note ends with a new Granville–Hardy–Littlewood style conjecture predicting

$$\log P(N) \sim \frac{(\log N)^2}{\log \log N}.$$

1 The problem

For $N \geq 1$, write $[N] = \{1, 2, \dots, N\}$. A family $\mathcal{F} \subseteq 2^{[N]}$ is called *prime-intersecting* if for every two distinct sets $A, B \in \mathcal{F}$ the integer $|A \cap B|$ is prime. Let

$$P(N) = \max\{|\mathcal{F}| : \mathcal{F} \subseteq 2^{[N]} \text{ is prime-intersecting}\}.$$

The question from the list is:

Is $P(N)$ polynomial in N , or can it be exponential?

Here “polynomial” means $P(N) \leq N^C$ for some absolute constant C , and “exponential” means $P(N) \geq 2^{cN}$ along all sufficiently large N or at least along an infinite sequence, for some fixed $c > 0$. The answer is neither:

$$P(N) \text{ grows faster than every fixed power of } N, \quad P(N) = 2^{o(N)}.$$

2 Main result

All logarithms below are natural unless the base is explicitly shown. Define the tower functions by

$$\text{twr}_1(x) = x, \quad \text{twr}_{j+1}(x) = 2^{\text{twr}_j(x)}.$$

Thus $\log_2^{(7)} N$ denotes the inverse operation to a height-seven base-2 tower, up to harmless additive constants.

Theorem 2.1 (Unconditional bounds). *There is an absolute constant $c > 0$ such that, as $N \rightarrow \infty$,*

$$N^{c \log_2^{(7)} N} \leq P(N) \leq \exp\left(\left(\frac{1}{2} + o(1)\right) \frac{N \log \log N}{\log N}\right).$$

In particular $P(N)$ is superpolynomial but subexponential.

The lower bound uses the Green–Tao theorem and its known quantitative extraction. If one only wants the qualitative conclusion “not polynomial”, the ordinary Green–Tao theorem is enough: for every fixed r , it gives $P(N) \gg_r N^r$.

3 Lower bounds from arithmetic progressions of primes

The key construction turns an arithmetic progression of primes into a set system. It is simple but surprisingly powerful.

Lemma 3.1 (Prime progression block construction). *Suppose*

$$a, a + d, a + 2d, \dots, a + (r - 1)d$$

is an arithmetic progression of r primes. If m is a positive integer with

$$a + dm \leq N,$$

then

$$P(N) \geq \binom{m}{r}.$$

Proof. Choose pairwise disjoint sets

$$C, B_1, B_2, \dots, B_m \subseteq [N]$$

with

$$|C| = a, \quad |B_i| = d \quad (1 \leq i \leq m).$$

This is possible because $a + dm \leq N$. For every r -element subset $X \subseteq [m]$, define

$$A_X = C \cup \bigcup_{i \in X} B_i.$$

Let

$$\mathcal{F} = \{A_X : X \in \binom{[m]}{r}\}.$$

If $X \neq Y$, then $|X \cap Y| \in \{0, 1, \dots, r - 1\}$, and hence

$$|A_X \cap A_Y| = |C| + d|X \cap Y| = a + d|X \cap Y|.$$

This is one of the primes $a, a + d, \dots, a + (r - 1)d$. Therefore $\mathcal{F}$ is prime-intersecting and has size $\binom{m}{r}$. $\square$

Corollary 3.2 (Superpolynomial growth). *For every fixed $r \geq 1$ there is a constant $c_r > 0$ such that*

$$P(N) \geq c_r N^r$$

for all sufficiently large N . Hence $P(N)$ is not polynomially bounded.

Proof. By Green and Tao's theorem, the primes contain an arithmetic progression of length r . Fix such a progression $a, a + d, \dots, a + (r - 1)d$. Taking

$$m = \left\lfloor \frac{N - a}{d} \right\rfloor$$

in Lemma 3.1 gives $m \gg_r N$, and so

$$P(N) \geq \binom{m}{r} \gg_r N^r.$$

□

For a quantitative growing exponent one needs a quantitative version of Green–Tao. Green and Tao extracted from their proof the fact that the first k -term arithmetic progression of primes has all terms bounded by an exponential tower in k of height seven.

Corollary 3.3 (Explicit but very slow lower bound). *There is an absolute constant $c > 0$ such that*

$$P(N) \geq N^{c \log_2^{(7)} N}$$

for all sufficiently large N .

Proof. The quantitative Green–Tao bound says that there is an absolute constant C such that, for every r , there is an r -term arithmetic progression of primes whose largest term is at most $\text{twr}_7(Cr)$.

Choose

$$r = \left\lfloor c_0 \log_2^{(7)} N \right\rfloor$$

with $c_0 > 0$ sufficiently small. Then, for all large N , there is an r -term prime progression

$$a, a + d, \dots, a + (r - 1)d$$

with largest term at most $N^{1/2}$. In particular $a \leq N^{1/2}$ and $d \leq N^{1/2}$. Hence

$$m = \left\lfloor \frac{N - a}{d} \right\rfloor \geq \frac{1}{2} N^{1/2}$$

for large N . Lemma 3.1 gives

$$P(N) \geq \binom{m}{r}.$$

Since $r = o(\log N)$,

$$\log \binom{m}{r} \geq r \log \frac{m}{r} \geq \left(\frac{1}{2} + o(1) \right) r \log N.$$

Substituting $r \asymp \log_2^{(7)} N$ proves the claim, after decreasing the absolute constant c if necessary. □

4 The upper bound

The upper bound is a restricted-intersection argument. The important point is that within a fixed uniform layer, not every prime up to N can occur as an intersection size.

For $0 \leq k \leq N$, let

$$\mathcal{F}_k = \{A \in \mathcal{F} : |A| = k\}.$$

If $A, B \in \binom{[N]}{k}$ are distinct, then

$$\max(0, 2k - N) \leq |A \cap B| \leq k - 1.$$

Therefore, inside the k th layer, the only possible prime intersection sizes lie in

$$I_k = [\max(0, 2k - N), k - 1].$$

Let

$$s_k = |I_k \cap \mathbb{P}|,$$

where $\mathbb{P}$ denotes the set of primes.

Lemma 4.1 (Polynomial method for one layer). *For every k ,*

$$|\mathcal{F}_k| \leq \sum_{j=0}^{s_k} \binom{N}{j}.$$

Proof. Let

$$L_k = I_k \cap \mathbb{P}.$$

For each $A \in \mathcal{F}_k$, define a polynomial in N variables by

$$f_A(x_1, \dots, x_N) = \prod_{p \in L_k} \left(\sum_{i \in A} x_i - p \right).$$

Restrict these polynomials to the Boolean cube $\{0, 1\}^N$, reducing by $x_i^2 = x_i$ so that every polynomial becomes multilinear. The degree is at most s_k .

For $B \in \mathcal{F}_k$, write $\mathbf{1}_B$ for its incidence vector. If $B \neq A$, then $|A \cap B| \in L_k$, so

$$f_A(\mathbf{1}_B) = 0.$$

On the other hand,

$$f_A(\mathbf{1}_A) = \prod_{p \in L_k} (k - p) \neq 0,$$

because every $p \in L_k$ satisfies $p \leq k - 1$.

Thus the functions f_A are linearly independent on the Boolean cube: if $\sum_A c_A f_A = 0$, evaluating at $\mathbf{1}_B$ leaves only the term $c_B f_B(\mathbf{1}_B)$, so $c_B = 0$. The vector space of multilinear polynomials of degree at most s_k in N variables has dimension

$$\sum_{j=0}^{s_k} \binom{N}{j}.$$

This proves the lemma. □

It remains to estimate s_k uniformly in k .

Lemma 4.2 (How many primes can occur in one layer). *As $N \rightarrow \infty$,*

$$\max_{0 \leq k \leq N} s_k \leq \left(\frac{1}{2} + o(1) \right) \frac{N}{\log N}.$$

Proof. If $k \leq N/2$, then

$$s_k \leq \pi(N/2) = \left(\frac{1}{2} + o(1) \right) \frac{N}{\log N}$$

by the prime number theorem.

Now suppose $k > N/2$ and put $h = N - k$. Then $0 < h < N/2$ and

$$I_k = [N - 2h, N - h - 1],$$

with the lower endpoint truncated at 0 if necessary. Thus I_k is an interval of length h .

If $h \leq N/4$, the Brun–Titchmarsh inequality in the form

$$\pi(x + h) - \pi(x) \leq (2 + o(1)) \frac{h}{\log h}$$

for intervals of length $h \rightarrow \infty$ gives

$$s_k \leq (2 + o(1)) \frac{h}{\log h} \leq \left(\frac{1}{2} + o(1) \right) \frac{N}{\log N},$$

since $h/\log h$ is increasing for large h and $h \leq N/4$. If h is bounded or very small, the trivial estimate $s_k \leq h$ is already $o(N/\log N)$.

It remains to consider $N/4 < h < N/2$. Write

$$u = N - h, \quad v = N - 2h.$$

If $v < N/\log^2 N$, then

$$u < N/2 + N/(2\log^2 N),$$

and hence

$$s_k \leq \pi(u) \leq \left(\frac{1}{2} + o(1) \right) \frac{N}{\log N}.$$

If instead $v \geq N/\log^2 N$, then the prime number theorem, uniformly for $v \leq x \leq N$, gives

$$s_k \leq \pi(u) - \pi(v) \leq \frac{u}{\log u} - \frac{v}{\log v} + o\left(\frac{N}{\log N}\right).$$

Since $u \geq v$ and $\log v \geq \log N - 2\log \log N$,

$$\frac{u}{\log u} - \frac{v}{\log v} \leq \frac{u - v}{\log v} = \frac{h}{\log v} \leq \left(\frac{1}{2} + o(1) \right) \frac{N}{\log N}.$$

This completes the proof. □

Proof of Theorem 2.1, upper bound. Let

$$S_N = \max_{0 \leq k \leq N} s_k.$$

By Lemmas 4.1 and 4.2,

$$|\mathcal{F}| = \sum_{k=0}^N |\mathcal{F}_k| \leq (N+1) \sum_{j=0}^{S_N} \binom{N}{j},$$

where

$$S_N \leq \left(\frac{1}{2} + o(1)\right) \frac{N}{\log N}.$$

Since $S_N = o(N)$,

$$\sum_{j=0}^{S_N} \binom{N}{j} \leq (S_N + 1) \left(\frac{eN}{S_N}\right)^{S_N}.$$

Therefore

$$\log |\mathcal{F}| \leq S_N \log \frac{eN}{S_N} + o\left(\frac{N \log \log N}{\log N}\right) \leq \left(\frac{1}{2} + o(1)\right) \frac{N \log \log N}{\log N}.$$

Taking the maximum over prime-intersecting $\mathcal{F}$ proves

$$P(N) \leq \exp\left(\left(\frac{1}{2} + o(1)\right) \frac{N \log \log N}{\log N}\right).$$

Together with Corollary 3.3, this proves Theorem 2.1. $\square$

5 A Granville–Hardy–Littlewood conjecture

The proved bounds leave a large gap:

$$\exp(c \log N \cdot \log_2^{(7)} N) \leq P(N) \leq \exp\left(\left(\frac{1}{2} + o(1)\right) \frac{N \log \log N}{\log N}\right).$$

The lower bound is limited by the currently available quantitative Green–Tao bounds. Prime-pattern heuristics suggest a much larger lower bound and lead to a natural conjecture for the true order of magnitude.

Granville’s analysis of prime patterns predicts that the smallest r -term arithmetic progression of primes has largest prime roughly

$$\left(e^{1-\gamma} \frac{r}{2}\right)^{r/2},$$

where γ is Euler’s constant. He also derives, from the quantitative prime k -tuple heuristic, the estimate that the number of r -term arithmetic progressions of primes with all terms at most x should be about

$$\left(\frac{e^{\gamma-1} \log r}{\log x}\right)^r x^2,$$

up to lower-order factors. For the block construction, however, the common difference matters as much as the largest prime: a small common difference produces many blocks.

Let

$$W_r = \prod_{p \leq r} p.$$

The difference $d = W_r$ is the smallest natural “fully admissible” choice: if $p \leq r$ and $p \nmid d$, then the residues

$$a, a + d, \dots, a + (r - 1)d \pmod{p}$$

cover all residue classes modulo p , forcing one term to be divisible by p . Taking $d = W_r$ avoids this local obstruction.

The Hardy–Littlewood heuristic for the one-variable pattern

$$a, a + W_r, \dots, a + (r - 1)W_r$$

predicts roughly

$$\frac{N}{(\log N)^r} \mathfrak{S}_r(W_r)$$

valid starting values $a \leq N - (r - 1)W_r$, where $\mathfrak{S}_r(W_r)$ is the singular series. For $p \leq r$, the local factor is

$$(1 - 1/p)^{1-r},$$

and Mertens’ theorem gives

$$\prod_{p \leq r} (1 - 1/p)^{1-r} = (e^\gamma \log r)^{r+o(r)}.$$

The primes $p > r$ contribute only lower-order exponential factors at this scale. Thus the logarithm of the expected count is

$$\log N - r \log \log N + r \log \log r + O(r).$$

This is large and positive for

$$r \leq (1 - o(1)) \frac{\log N}{\log \log N}.$$

For such r one expects a prime progression of length r and common difference W_r , with

$$W_r = \exp((1 + o(1))r) = N^{o(1)}.$$

The block construction would then give

$$m \sim \frac{N}{W_r} = N^{1-o(1)},$$

and hence

$$\log P(N) \geq \log \binom{m}{r} = (1 + o(1)) \frac{(\log N)^2}{\log \log N}.$$

This motivates the following conjecture.

Conjecture 5.1 (Granville–Hardy–Littlewood lower-bound conjecture). *For every fixed $\varepsilon > 0$ and all sufficiently large N , there exists an integer*

$$r \geq (1 - \varepsilon) \frac{\log N}{\log \log N}$$

such that the primes contain an r -term arithmetic progression

$$a, a + W_r, \dots, a + (r - 1)W_r$$

with $a + (r - 1)W_r \leq N$. Consequently,

$$P(N) \geq \exp \left((1 - o(1)) \frac{(\log N)^2}{\log \log N} \right).$$

The following stronger conjecture is the proposed revised form of Problem 6.

Conjecture 5.2 (Prime-intersection growth conjecture). *The true growth of $P(N)$ is quasi-polynomial on the logarithmic scale:*

$$\log P(N) \sim \frac{(\log N)^2}{\log \log N}.$$

Equivalently,

$$P(N) = \exp\left((1 + o(1)) \frac{(\log N)^2}{\log \log N}\right).$$

Remark 5.3. Conjecture 5.2 is deliberately stronger than what current methods prove. The lower-bound side is a direct Hardy–Littlewood prediction for the same block construction used in Lemma 3.1. The upper-bound side would require new ideas beyond the present Ray–Chaudhuri–Wilson polynomial method, which only gives the much weaker subexponential upper bound in Theorem 2.1.

6 Summary

Problem 6 asked whether $P(N)$ is polynomial or can be exponential. The answer is:

$$P(N) \text{ is superpolynomial but subexponential.}$$

The proof has two independent parts.

1. Arithmetic progressions of primes give block families. Qualitative Green–Tao implies $P(N) \geq c_r N^r$ for every fixed r , and the quantitative Green–Tao tower bound gives $P(N) \geq N^{c \log_2^{(7)} N}$.
2. The polynomial method applied separately to each uniform layer gives

$$P(N) \leq \exp\left(\left(\frac{1}{2} + o(1)\right) \frac{N \log \log N}{\log N}\right).$$

The proposed next target is Conjecture 5.2: the expected true order is

$$\log P(N) \sim \frac{(\log N)^2}{\log \log N},$$

with the lower-bound heuristic coming from prime progressions with primorial common difference.

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